

Introduction to Biostatistics /Stat2091

Course Title /Code	Introduction to Biostatistics / Stat2091				
Degree Program	BSc. in Human Nutrition				
Module Title/Code	Biostatistics and Research Methodology /HuNu-M3091				
Course Information	Year: II Semester: II				
Credit Points	5 CP				
Student Work Load in Hours/Week	Lecture	Tutorial	Lab practice	Home study	Total
	3	2	0	3	8
Course Description/ Content	This course is designed to equip the students with knowhow and skills of biostatistics. The main core contents are; introduction to statistics, Methods of Data collection, Organization and presentation, summarizing data, Demographic methods and health service statistics, Probability, Sampling methods, Basic concepts of estimations, Hypothesis testing, Correlation and regression. For each of the biostatistics concepts examples are drawn from nutrition and other health sciences statistical analysis using computer programmed.				
Course Objective and Competences to be Acquired	At the end of the course, students will be able to: Define Statistics and Biostatistics Enumerate the importance and limitations of statistics Define and Identify the different types of data and understand why we need to classifying variables understand the methods of data collection, organization, presentation, analysis and interpretation; apply statistical methods in scientific researches, decision making and future career to solve standard problems in their field of study and in a wide range of disciplines; communicate or present the results of their analyses in clear and non-technical language; discuss biases and confounding scenario in data collected undertake research projects and interpreting epidemiological information use standard statistical techniques and computer applications to analyze epidemiological data				
Pre-requisite(s)	None				
Course status	Compulsory				

Teaching and learning methods: Variety of instructional methods including lectures, tutorials, discussion, group work, assignments and presentation and individual learning are involved.

Schedule

Weeks	Topics and Sub Topics	Required Text
1 st	Introduction statistics Definition and classification of statistics Rationale of studying statistics Stages in statistic investigation Definitions of some basic terms Applications, uses and limitation of statistics Types of variables and measurement scales Methods of Data Collection and Presentation	
2 nd	Methods of data collection Sources of data Methods of collection Methods of data presentation Motivating examples Frequency distributions: qualitative, quantitative: absolute, relative and percentage Tabular presentation of data Diagrammatic presentation of data: bar charts, pie-chart, cartograms Graphical presentation of data: histogram, and frequency polygon Measures of Central Tendency Motivating examples Objectives of measures of central tendency Summation notation Important characteristics of good average Mean Arithmetic mean Geometric mean Harmonic mean Median Mode	
3 rd	Measures of Variation (Dispersion), Skewness and Kurtosis Motivating examples Objectives of measures of central tendency Measure of dispersion/variation Range Variance, standard deviation and coefficient of variation	

4-5th 6-7th 8-9th 10th	Standard scores Skewness Kurtosis Demographic Methods and Health Services Statistics Introduction Sources of demographic data Stages in demographic transition Vital Statistics Measures of Fertility Measures of Mortality Population growth and Projection Health services statistics Elementary Probability and Probability Distributions Introduction Definition and some concepts (experiment, sample, event, equally likely outcomes, mutually exclusive events, independent events) Random experiments Counting rules: addition, multiplication, permutation, and combination rules Definitions of probability (probability of an event) Some rules of probability Random variables (discrete and continuous) and probability distributions Introduction to expectation: mean and variance of random variable Common discrete distribution: binomial and Poisson Common continuous distributions: Normal, t and chi-square distributions Sampling and Sampling Distribution of Mean Basic concepts (population, sample, parameter, static, sample frame, sampling unit, sampling error, sample size) Reasons for sampling Types of sampling (probability and non-probability sampling) Simple random sampling (lottery, table or computer generated random numbers) Sampling distribution of the sample mean and proportions Central limit theorem Estimation and Hypothesis Testing Estimation Motivating examples Point estimation: mean and proportion	
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11-13 th 14-15 th 16 th	Interval estimation: mean and proportion Hypothesis testing Motivating examples Important concepts in testing a statistical hypothesis Steps in testing hypothesis Hypothesis testing about the population mean Hypothesis testing about the population proportion Chi-square test of association Simple Linear Regression and Correlation Motivating examples Introduction: regression and correlation Simple linear regression Correlation coefficient Final Exam	
Assessment/ Evaluation	Assignment and presentation: - 30% Quiz 1:- 5% Quiz 2:- 5% Test: - 10% Final exam: - 50%	
References	Coolidge, F.L. (2006). Statistics: A Gentle Introduction (2 nd edition). Gupta, C.B. and Gupta, V. (2004). An Introduction to Statistical Methods. Vikas Publishing House, Pvt. Ltd, India	